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Computational analysis of yield stress buildup and stability of deposited layers in material extrusion additive manufacturing

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ABSTRACT

This paper investigates the stability of deformable layers produced by material extrusion additive manufacturing. A Computational Fluid Dynamics (CFD) model is developed to predict the deposition flow of viscoplastic materials such as ceramic pastes, thermosets, and concrete. The viscoplastic materials are modelled with the Bingham rheological equations and implemented with a generalized Newtonian fluid model. The developed CFD model applies a scalar approach to differentiate the rheology of two layers in order to capture the deposition of a wet layer onto a semi solidified printed layer (i.e., wet-on-semisolid printing). The semi solidification is modelled by a yield stress buildup. The cross-sectional shapes of the deposited layers are predicted, and the relative deformation of the first layer is studied for different yield stress buildups and processing conditions such as printing- and extrusion-speed, layer height, and nozzle diameter. The results of the CFD model illustrate that the relative deformation of the first layer decreases non-linearly with an increase in yield stress, and that stable prints can be obtained when taking into account the semi solidification. Furthermore, it is found that the deformation is dependent on a non-trivial interplay between the extrusion pressure, the shape of the cross-section, and the contact area between the layers. Finally, the results highlight which process conditions can be changed with benefit in order to limit the requirement on the yield stress buildup and still provide stable prints.

1. Introduction

Three-dimensional fabrication of structures produced by Material Extrusion Additive Manufacturing (MEX-AM) is an attractive technology for a wide range of industries [1–3]. MEX-AM is an umbrella term that covers multidisciplinary fabrication techniques such as “Fused Deposition Modelling (FDM) or Fused Filament Fabrication (FFF)”[4], “Robocasting (RC)” [5], and “3D Concrete Printing (3DCP)” [6]. Thermoplastics have been a commonly used material in MEX-AM, but recently viscoplastic materials such as ceramics [7], thermosets [8], and concrete [9] have received attention. Typically, these materials are used in RC and 3DCP, and are extruded through a nozzle and deposited onto a solid substrate following a layer-by-layer approach [10]. Control over the material properties is crucial when attempting to retain the layered shape of deposited viscoplastic materials, as the layered shape can be flattened due to the applied pressure (from subsequent layers) when the material is too liquid, while a too rigid material rarely flows through the nozzle.

During MEX-AM of viscoplastic materials, a printed layer can be considered as wet, semisolid, or solidified, depending on the material state. Depositing a new layer onto a solidified layer typically results in a geometrically stable print (i.e., homogeneous width and height of layers), but it is often not ideal from a printing time perspective (if the solidification time is long) and it can be detrimental for the interlayer bonding [11]. Consequently, printing on a wet or semisolid layer is in many cases preferable. However, in this case, printing geometrically stable layers is not a simple task [12,13], as discussed in several experimental studies on MEX-AM of viscoplastic materials [14–17]. Different studies showed that all layers potentially could deform during the deposition, however the most critical deformation takes place at the first layer [18–20].

Computational Fluid Dynamics (CFD) modelling of extrusion-deposition flows has been found very advantageous when analyzing experimental MEX-AM results [21,22]. In regards to MEX-AM with

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thermoplastic, Xia et al. [23] estimated the geometrical shape of the deposited layers using numerical simulations. In addition, Comminal et al. [24,25] and Mollah et al. [26] developed CFD models with the aim to predict and discuss precision of printed corners, while Serdeczny et al. [27,28] addressed through numerical modelling how to reduce porosities and enhance the bonding between subsequent layers. Numerical results have also pointed out that a deposited layer undergoes two pressure contributions: the hydrostatic pressure due to the layers weight and the extrusion pressure applied by the extruded material [29]. Comminal et al. [22,30] explained the effect of extrusion pressure on the geometry of a single layer. It was seen that the extrusion pressure facilitates sideways flow of the depositing material. During printing multiple layers, the extrusion pressure can affect the geometrical shape of the previously printed layers [31], although it depends on the material state of printed layers.

When printing on top of a wet layer that has a viscoplastic behavior (i.e., wet-on-wet), the previously printed layers undergo deformations due to the combined effect of hydrostatic- and extrusion-pressure, as reported by Mollah et al. [32]. The study concluded that the deformation could be reduced to some extent by changing the properties of the material (yield stress and plastic viscosity) and processing conditions (extrusion and printing speeds, layer height, and nozzle diameter). However, it was found non-trivial to fully eliminate the deformation. Furthermore, the study simulated printing on top of a solid layer and showed, unlike wet-on-wet printing, that the extrusion pressure values of different layers (first to fifth) are constant as no deformation occurred. This gave a conservative estimate of the required yield stress of the printed layer in order not to deform. The conservative estimate was that the yield stress of the printed layer must be higher than the total applied pressure (i.e., the sum of hydrostatic- and extrusion-pressure). If the total applied pressure is high, the yield stress buildup is required to be high as well. However, obtaining a high buildup could be time-consuming if the solidification time is long and it can lead to reduced interlayer bonding strength [11]. Consequently, it is crucial to understand the effect of yield stress buildup on the geometrical stability of layers, i.e., wet-on-semisolid printing (which typically is the case for MEX-AM with ceramics, thermosets and concrete).

This paper presents the first CFD model of two-layered viscoplastic prints by MEX-AM that accounts for the time dependent material development during the process (i.e., the second layer is deposited onto a semisolid first layer). The semi solidification is modelled by increasing the yield stress of the first layer while the second layer is deposited; thereby mimicking a yield stress buildup of the material. The model is exploited to investigate geometrical stability of prints performed with materials with various yield stress buildup under a wide range of process conditions. The outline of the work is as follows. In Section 2, details about the CFD model are presented. The simulated results predicting the influence of material development and processing parameters on geometrical stability are discussed in Section 2.2. Finally, conclusions are summarized in Section 3.

2. Methodology

2.1. Computational model

The CFD model presented in this study is an extension of a model developed to simulate MEX-AM with non-Newtonian materials, which previously was validated by comparison with single and multi-layer experimental results [30,31]. The extension consists in adding a time dependency to the material behavior, which is typically seen for concrete printing due to its hydration process and for thermoset printing due to the cross-linking of polymer chains. Details about this time dependent material modelling is given in section 2.2. The CFD model simulates the deposition flow of the time-dependent viscoplastic material inside a computational domain of $10D \times 1.5D \times (1.5D + hN)$, where D is the diameter of the cylindrical nozzle, N is the number of layers, and h is the distance between the solid substrate and the nozzle orifice (i.e., the nominal height of the first layer), see Fig. 1. The nozzle extrudes the material onto the substrate in a layer-by-layer manner. A constant extrusion speed U is applied to the inlet that evolves into a fully developed profile at the orifice of the nozzle. When the material touches the substrate, the nozzle starts to travel along a straight toolpath with a prescribed printing speed V. The nozzle travels a distance of $8D$ while printing the layers. The second layer is also printed with the distance h between the nozzle orifice and the first layer.

A uniform Cartesian grid is used to mesh the computational domain. A mesh sensitivity analysis was performed with the cell sizes 0.8, 0.9, and 1.0 mm, which resulted in 0.58, 0.78, and 1.1 million cells, respectively. A cell size of 0.9 mm was chosen as it was found to be time-efficient and had a negligible effect on the accuracy of the results. The top plane of the computational domain is treated as an inlet boundary, however a solid surface is superimposed to it. This permits the material to enter the domain only through the nozzle orifice that is attached to the top plane. A stationary wall condition is assigned to the bottom plane that carries the solid substrate. In the xz-plane, a symmetry boundary is applied to reduce the computational cost. Other planes are assigned with continuative boundary conditions that allow the fluid to freely exit the computational domain, although the deposited material does not interact with those planes. The solid (i.e., nozzle and substrate) and fluid interaction is prescribed with a no-slip boundary condition.

2.2. Rheology of layers

In this study, the viscoplastic constitutive model adapted by the CFD model is the Bingham material model [33]:

$$\tau = \eta_P \dot{\gamma} + \tau_{printing} \tag{1}$$

where τ is the shear stress, η_P is the plastic viscosity, $\dot{\gamma}$ is the shear rate, and $\tau_{printing}$ is the yield stress of the material while being deposited. The latter is typically represented by τ_0, but the material changes yield stress over time as described in detail later in the section and for that reason this nomenclature is chosen. When the stress state in the material exceeds the yield stress, the fluid starts flowing. The apparent viscosity function describing the Bingham material model is given by:

$$\mu_{app}(\dot{\gamma}) = \eta_P + \tau_{printing}/\dot{\gamma} \tag{2}$$

This becomes infinite at zero shear rate, so in order to avoid the numerical singularity, a bi-viscous regularization [34] of the Bingham material model is used in the CFD model:

$$\mu_{app}(\dot{\gamma}) = \begin{cases} \mu_{\max.app}, & \text{when } \dot{\gamma} < \dot{\gamma}_c \\ \eta_P + \dfrac{\tau_{printing}}{\dot{\gamma}}, & \text{when } \dot{\gamma} \geq \dot{\gamma}_c \end{cases} \tag{3}$$

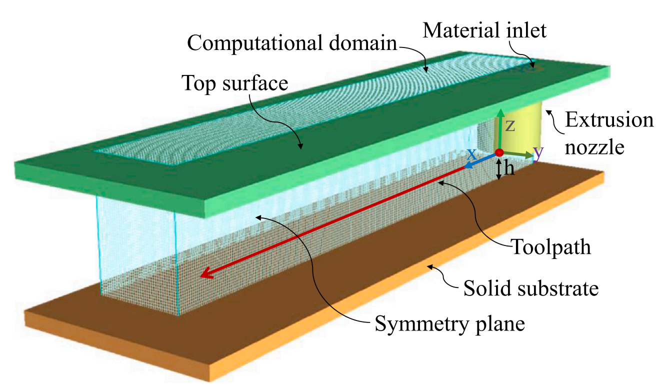

Fig. 1. Geometry with toolpath, computational domain, and boundary conditions of the CFD model.

where $\mu_{\max.app} = \eta_P + \tau_{printing}/\dot{\gamma}_c$ is the maximum apparent viscosity and $\dot{\gamma}_c$ is the critical shear rate. The selection of $\dot{\gamma}_c$ should be small enough (i.e., resulting in a sufficiently larger maximum apparent viscosity) to limit any flow during the simulation time in the parts of the domain where the stress state is lower than the yield stress, while large enough to avoid extensive computational time and possible divergence issues. The value of $\dot{\gamma}_c$ is selected in this study based on the following two analyses on the cross-section of a single layer. One compared the cross-sections for different values of $\dot{\gamma}_c$ (0.03, 0.05, and 0.10 s^{-1}). The other analysis compared the cross-sections at different times for which the simulated layer was kept at rest. No significant change was observed. Therefore, $\dot{\gamma}_c = 0.05\ s^{-1}$ is used in all simulations.

As previously mentioned, the CFD model takes into account that the viscoplastic material changes its rheological behavior over time. Once the first layer is deposited, its yield stress is changed before the second layer is printed on top (i.e., the first layer is now assumed semisolid). Here it should be noted that the second layer is printed with the initial rheological properties of the first layer. The change in yield stress of the first layer is modelled with a scalar approach. This is done by assigning a scalar S_c in terms of mass per fluid volume, which is larger than zero, exclusively to the first layer after deposition with an accompanying apparent viscosity function:

$$\mu_{SC.app}(\dot{\gamma}) = \begin{cases} \mu_{\max.SC.app}, & \text{when } \dot{\gamma} < \dot{\gamma}_c \\ \eta_P + \dfrac{\tau_{printed}}{\dot{\gamma}}, & \text{when } \dot{\gamma} \geq \dot{\gamma}_c \end{cases} \tag{4}$$

where $\mu_{\max.SC.app} = \eta_P + \tau_{printed}/\dot{\gamma}_c$. The only difference as compared to Eq. 1 is $\tau_{printed}$ that represent the yield stress of the already deposited material, whereas $\tau_{printing}$ is the yield stress of the material while being deposited. Thus, two apparent viscosity functions are assigned to the first layer, one from the material itself (Eq. 3) and the other from the scalar (Eq. 4). They are governed with a viscosity mixture-rule that uses a mass-average between the apparent viscosities of both material and scalar [35]:

$$\mu_{mix} = \frac{\rho\mu_{app} + S_c\mu_{SC.app}}{\rho + S_c} \tag{5}$$

where ρ is the density. This means, in the part of the domain where $S_c = 0$, μ_{mix} is equal to μ_{app}, while in the part of the domain where $S_c > 0$, μ_{mix} is larger than μ_{app}. However, the increased viscosity (when $S_c > 0$) does not comply with the rheological behavior of the first layer after being deposited. Therefore, μ_{mix} is used as a classification parameter in Eq. (4) that specifies whether μ_{app} or $\mu_{SC.app}$ is used as the apparent viscosity in a given control volume:

$$\mu_{cv.app}(\dot{\gamma}) = \begin{cases} \mu_{SC.app}, & \text{when } \mu_{mix} > \mu_{app} \\ \mu_{app}, & \text{when } \mu_{mix} \leq \mu_{app} \end{cases} \tag{6}$$

With this procedure, wet-on-wet printing (i.e., $\tau_{printed} = \tau_{printing}$) is simulated by specifying $S_c = 0$, while wet-on-semisolid printing (i.e., $\tau_{printed} > \tau_{printing}$) is achieved by given S_c an arbitrary value larger than zero.

2.3. Governing equations

The transient and isothermal flow dynamics of the incompressible, viscoplastic material are modelled by the continuity and momentum equations:

$$\nabla . \boldsymbol{u} = 0 \tag{7}$$

$$\rho\left(\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u}.\nabla\boldsymbol{u}\right) = \rho\boldsymbol{g} - \nabla p + \nabla . \boldsymbol{S}_T \tag{8}$$

where $\boldsymbol{u}$ is the velocity vector, $\boldsymbol{g}$ is the gravitational acceleration vector with components $(0, 0, -g)$, p is the pressure, t is the time, and $\boldsymbol{S}_T$ the deviatoric stress-tensor. The apparent viscosity function is seen in the constitutive stress-tensor, which is modelled as:

$$\boldsymbol{S}_T = 2\mu_{cv.app}(\dot{\gamma})\boldsymbol{D}_T \tag{9}$$

where $\dot{\gamma} = \sqrt{2\mathrm{tr}(\boldsymbol{D}_T^2)}$ represents the strain rate magnitude and $\boldsymbol{D}_T$ represents the deformation rate tensor, defined as:

$$\boldsymbol{D}_T = \frac{1}{2}\left((\nabla\boldsymbol{u}) + (\nabla\boldsymbol{u})^T\right) \tag{10}$$

where the superscript T represents the transpose notation.

2.4. Simulation framework

The CFD model is developed in the commercial software ***FLOW-3D***®, version 12.0 [35]. The model solves a single fluid with a free surface. The free surface tracking is done by one of the Volume-of-Fluid (VOF) methods, the Split Lagrangian method known as TruVOF, see details in [37,38]. The governing equations are discretized by the Finite Volume Method (FVM) and the viscoplastic behavior is predicted by the built-in generalized Newtonian fluid framework. The viscosity is calculated in a separate subroutine that has been customized according to section 2.2. An implicit technique, successive under-relaxation, is used to solve the viscous stress of the momentum equation. The momentum advection is calculated explicitly by a monotonically preserving upwind-difference technique that ensures second-order accuracy in space and first-order accuracy in time. The pressure and velocity components are solved implicitly in time. The predictions of pressure and forces near walls are facilitated by the immersed boundary method, which assists in the prediction of the extrusion pressure [36]. The time step size is initially set to a value of $1e^{-5}$, however it is dynamically determined by the solver accordingly to a stability criteria in order to avoid numerical instabilities [35]. The software uses the FAVOR technique (Fractional Area/Volume Obstacle Representation) to represent solid geometries within the computational domain, which assists in simulating the motion of the nozzle.

Several cases are simulated in this study. A description of the cases and the corresponding processing parameters are presented in Table 1. The hydrostatic pressure is calculated as $P_H = N\rho gh$ (cf. Table 1), where $g = 9.81\ ms^{-2}$. The material properties used in the study are presented in Table 2.

2.5. Data processing

The geometrical stability of the prints and extrusion pressure are analyzed by importing the fluid fraction and pressure field from the CFD software into MATLAB. Note that the fluid fraction is evaluated at a value of 0.5. The cross-sections of layers are extracted at the middle of the deposited layer, see Fig. 2. The relative plastic deformation of the first layer after printing both layers is calculated by:

$$D_B = \frac{BW_2 - BW_1}{BW_1} \times 100\% \tag{11}$$

where BW_1 is the width of the first layer at the end of its deposition and BW_2 is the width of first layer at the end of the second layer deposition. Thus, the D_B value is small for stable prints.

The extrusion pressure P_E represents both the static and dynamic pressure at the nozzle orifice. It is directly applied to the printing surface by the extruding material [29]. During the deposition of the first layer, the printing surface is the solid substrate, whereas for second layer, it is the already printed first layer. The extrusion pressure acts for a short time since the nozzle moves [32], however, it helps the sideway flow of the depositing material [30]. To calculate the extrusion pressure, the gauge pressure values are directly extracted from the cross-sectional

Table 1
Model descriptions with processing parameters and hydrostatic pressure.

Case ID	Nozzle diameter D (mm)	Layer height h (mm)	Gap ratio h/D	Extrusion speed U (mm/s)	Printing speed V (mm/s)	Speed ratio $S_r = V/U$	Hydrostatic pressure P_H (Pa) (for a single layer)
1	25	12.5	0.5	40	50	1.25	123
2	25	12.5	0.5	40	40	1	123
3	25	12.5	0.5	40	30	0.75	123
4	25	12.5	0.5	30	30	1	123
5	25	12.5	0.5	20	30	1.5	123
6	25	18.75	0.75	40	50	1.25	184
7	25	25	1.0	40	50	1.25	246
8	0.4	0.2	0.5	40	50	1.25	2
9	1.5	0.75	0.5	40	50	1.25	7
10	50	25	0.5	40	50	1.25	246

Table 2
Material properties.

Parameter	Symbol (Unit)	Value
Density	ρ (kg.m^{-3})	1000
Dynamic yield stress of printing layer (printing yield stress)	$\tau_{printing}$ (Pa)	100–300
Dynamic yield stress of printed layer (printed yield stress)	$\tau_{printed}$(Pa)	100–1500
Plastic viscosity	η_P(Pa.s)	0.01–33
Critical shear rate	$\dot{\gamma}_c$ (s^{-1})	0.05

slice taken at the nozzle exit. Then, a surface-average of the gauge pressure values is calculated at six different moments during the travel of the nozzle. Finally, an average of those six gauge pressure is taken to obtain P_E.

3. Results and discussions

This section investigates the influence of various parameters on the stability of the first layer when a new layer is printed above. The parameters include the change in rheological behavior (i.e., the yield stress buildup) and processing conditions (i.e., the extrusion speed, printing speed, layer height, and nozzle diameter). The simulated results are discussed in terms of cross-sections of the deposited layers, the relative deformation of the first layer, and the extrusion pressure during deposition.

3.1. Effect of yield stress buildup

The influence of time-dependent rheological properties, represented by the yield stress buildup, is seen in Figs. 3 and 4 for Case ID 1 (see Table 1), where $V = 50$ mm/s, $U = 40$ mm/s, and $\eta_P = 33$ Pa.s. Fig. 3 illustrates, as expected, that when printing wet-on-wet larger deformation in the first layer is observed as compared to printing wet-on-semisolid. This is due to the higher effective viscosity in the first layer when printing wet-on-semisolid. Furthermore, it is seen that almost no visible deformation in the first layer is obtained when printing with $\tau_{printed} = 3 \times \tau_{printing}$. In Fig. 4-a, the relative deformation of the first layer is quantified as a function of $\tau_{printed}$, and a non-linear decrease in D_B is observed. By far the largest improvement in geometrical stability is obtained by the initial increase in $\tau_{printed}$, however it is pronounced for $\tau_{printing} = 100$ Pa. This is attributed to the fact that the hydrostatic pressure exceeds $\tau_{printed}$ ($\leq$ 200 Pa), which means in these cases the deformation takes place during the whole print and not only while the extrusion pressure is active on a given point of the print (i.e., when $P_H + P_E > \tau_{printed}$). For all investigated $\tau_{printing}$, the D_B reduces to less than 1 % when $\tau_{printed} \geq 600$ Pa, which for many applications would be an acceptable precision. In this regard, one has to remember that having a too high $\tau_{printed}$ (i.e., wet-on-solid printing) can lead to reduced inter-layer bonding strength [11]. In addition, it can be adverse for some materials to have a very fast yield stress buildup, e.g., for concrete this can increase the risk of crack formation [39], while thermosets can be more prone to residual stresses and thermal warpage [40].

Fig. 4-b presents the extrusion pressure during printing of both layers for different values of $\tau_{printing}$ as a function of $\tau_{printed}$. For any $\tau_{printing}$, the first layer experiences a higher P_E than the second layer because it is printed onto a non-deformable solid substrate. In addition, it is to be noticed that the extrusion pressure experienced by the second layer varies with $\tau_{printed}$. This is due to the effective distance between the printing surface (first layer) and the nozzle exit increases when the first layer deforms. At higher values of $\tau_{printed}$, the extrusion pressure of the second layer seems to coincide with P_E of first layer, which is a consequence of the first layer increasingly starting to act like a solid substrate. In Fig. 4-b, it is seen that the extrusion pressure increases with increasing $\tau_{printing}$, which affect the relative deformation to also increase as seen in Fig. 4-a when comparing the three values of $\tau_{printing}$ at $\tau_{printed} = 300$ Pa. However, the correlation between increased extrusion pressure and increased relative deformation is not a general trend as seen for the data points when $\tau_{printed} \geq 600$ Pa. This is attributed to the precision of the modelling, and in this regard, it should be noted that the relative

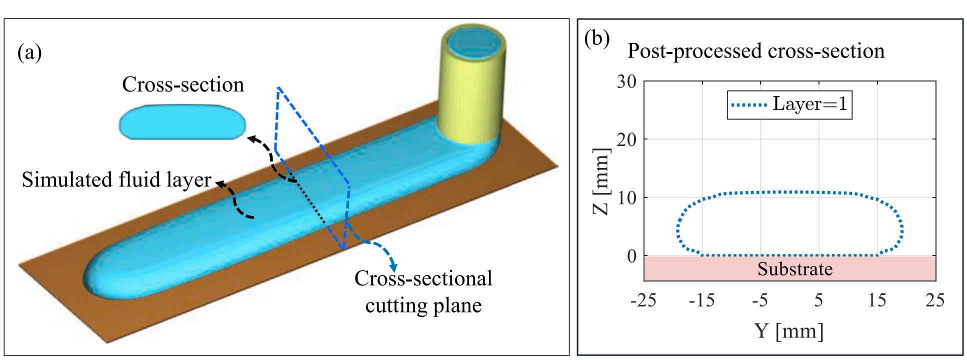

Fig. 2. Data processing; (a) Example of cross-sectional cutting plane that is used for evaluation, and (b) Post-processed result.

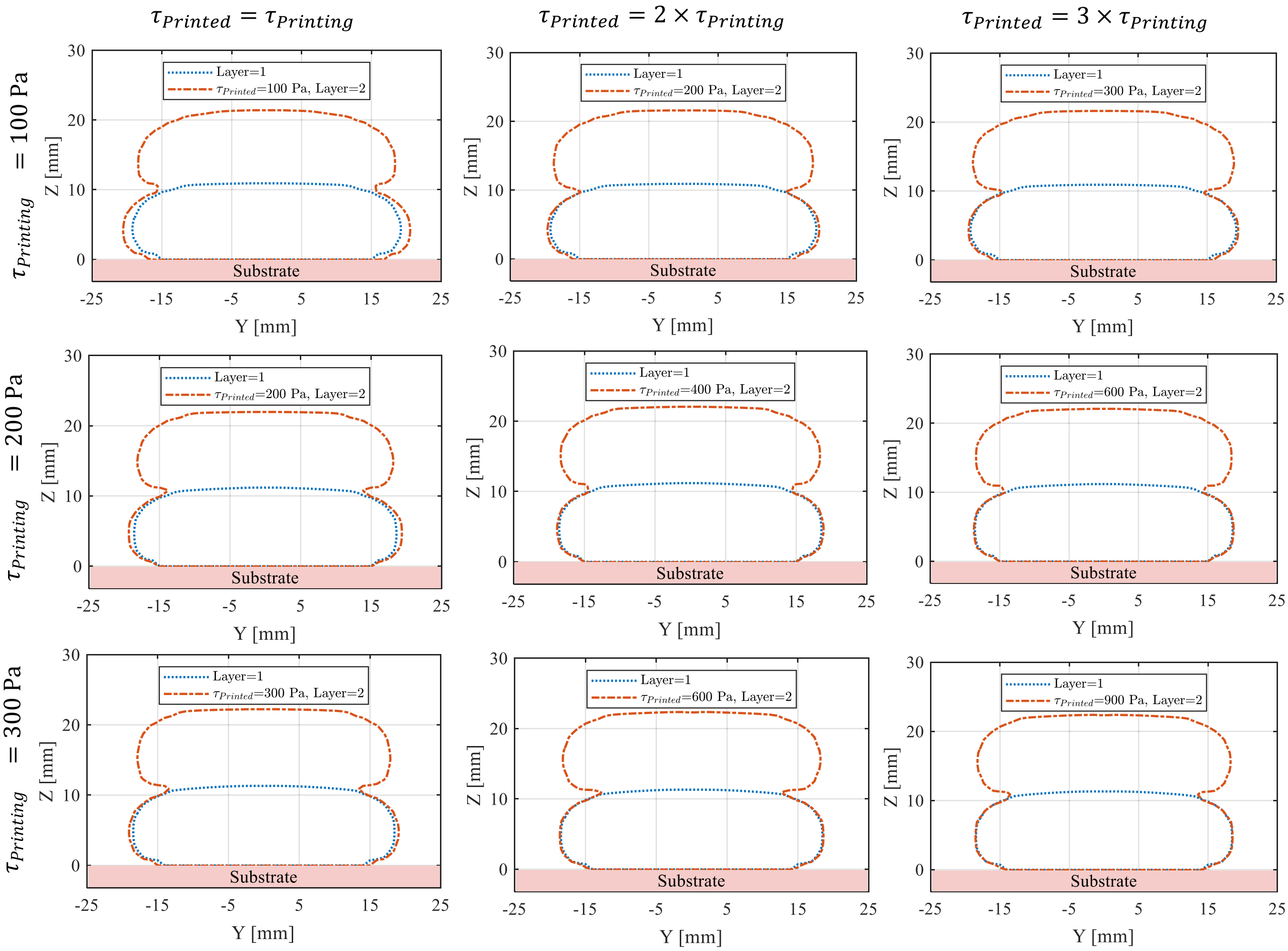

Fig. 3. Cross-sections of deposited layers for different printing yield stress $\tau_{printing}$ as well as printed yield stress $\tau_{printed}$.

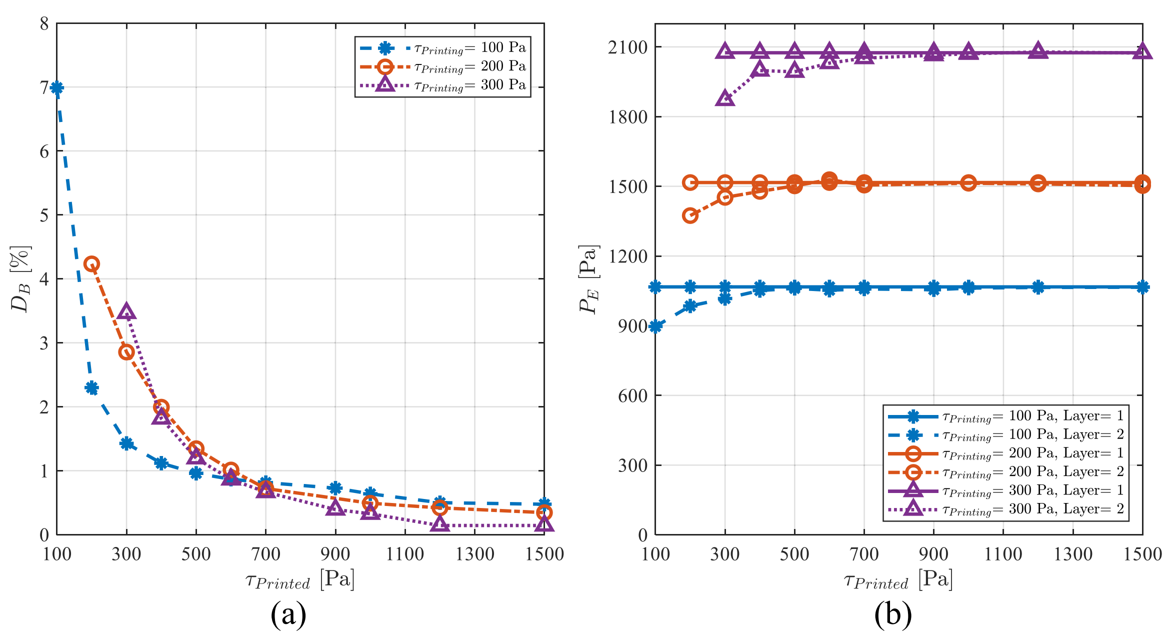

Fig. 4. (a) Relative deformation of first layer D_B and (b) extrusion pressure P_E as a function of printed yield stress $\tau_{printed}$, for different printing yield stress $\tau_{printing}$.

deformation is less than 1 %, which in absolute size represents a deformation of ~0.4 mm. Considering that the mesh cell size is around 1 mm, it is reasonable to doubt whether the results smaller than 1 % of relative deformation are accurate and reliable enough to draw any significant conclusions from.

3.2. Stability of layers in different processing conditions

This subsection investigates the stability of the first layer when considering variations in both processing conditions and the yield stress buildup of the first layer. Case IDs 1-10 (cf. Table 1) are simulated with $\tau_{printing} = 100$ Pa.

3.2.1. Printing speed

The printing speed's effect on geometrical stability is presented in Figs. 5 and 6 for Case IDs 1-3, where $U = 40$ mm/s and $\eta_P = 33$ Pa.s. Fig. 5 illustrates that both less material is deposited per unit length and less deformation is obtained in the first layer when increasing V. The drivers for the latter observation are: 1) less deposited material leads to lower effective layer height, which reduces the hydrostatic pressure and thereby the deformation; 2) higher printing speed leads to lower extrusion pressure (cf. Fig. 6-b) and less deformation; 3) higher printing speed means that the extrusion pressure will act for a shorter period of time on the first layer, which will reduce the deformation. These findings are in accordance with the results reported in [32].

Fig. 6-a shows a similar non-linear relationship between D_B and $\tau_{printed}$ as is seen in Fig. 4-a. For $\tau_{printed} \leq 200$ Pa, printing with a higher speed leads to substantially less deformation, but as the D_B values are relatively high for all printing speeds (and the deformation takes place due to the hydrostatic- and extrusion-pressure), it is unattractive for any application seeking geometrically stable layers. The figure also illustrates that strands printed with lower V converge faster to the smaller values of deformation D_B. This can be attributed to the higher initial deformation in those cases, especially in the range when the hydrostatic pressure is above the yield stress ($\tau_{printed} \leq 200$ Pa). At $\tau_{printed} = 600$ Pa, D_B becomes ~1 % for all printing speeds showing that for the investigated time-dependent viscoplastic materials no substantial gain in terms of geometrical stability is obtained by changing V.

3.2.2. Extrusion speed

The extrusion speed's effect on geometrical stability is presented in Figs. 7 and 8 for Case IDs 3-5, where $V = 30$ mm/s and $\eta_P = 33$ Pa.s. Fig. 7 shows that more material is deposited per unit length when increasing U, which in most cases lead to larger relative deformation in the first layer due to the larger effective layer height and increased extrusion pressure (see Fig. 8). The only exemption is for $U = 30$ mm/s at $\tau_{printed} = 200$ Pa, which is attributed to the fact that the results are plotted as a relative deformation and not an absolute deformation. Fig. 8-a shows, similarly as for Fig. 6-a, that the relative deformation decreases with increasing $\tau_{printed}$. However, the tendency of the combined effect of the extrusion speed and yield stress buildup on the deformation is opposite as compared to the tendency obtained in the relationship between the nozzle speed and yield stress buildup. For higher extrusion speeds, when the nozzle speed remains constant the deformation gets bigger, while for higher nozzle speeds, when the extrusion speed remains the same, the deformation gets smaller. Interestingly, considering that the speed ratio is defined as $S_r = V/U$, the effect of the combined speed ratio and yield stress buildup does not show a monotonic behavior. This means that in the case, for example, when $S_r = 1.25$ (cf. purple curve in Fig. 6-a) and $S_r = 1.5$ (cf. blue curve in Fig. 8-a) for $\tau_{printed} = 100$ Pa the deformation is similar (around 7 %), while for $\tau_{printed} = 200$ Pa, unexpectedly a smaller deformation is obtained for the lower speed ratio of 1.25. This indicates that both speeds have an independent effect on the deformation when combined with the yield stress buildup of the already printed strand. The anomalous relative deformation value for $U = 20$ mm/s and $U = 30$ mm/s when $\tau_{printed} = 200$ Pa can be attributed to an unexpectedly high deformation when $U = 20$ mm/s. Such high value can be justified with the rounder shape of the layer, which will be discussed in depth in the next section. Fig. 8-a also shows that for the investigated material- and process-parameters stable prints (i.e., $D_B = \sim 1$ %) is first achieved when $\tau_{printed} = 600$ Pa, and at this stage the extrusion pressure has a relatively limited effect on the geometrical stability.

3.2.3. Layer height

The layer height's effect on geometrical stability is presented in Figs. 9 and 10. The simulations include $h = 12.5$ mm (Case ID 1), $h = 18.75$ mm (Case ID 6), and $h = 25$ mm (Case ID 7) (cf. Table 1), where $U = 40$ mm/s, $V = 50$ mm/s, and $\eta_P = 33$ Pa.s are constant. Fig. 9 shows that by increasing the layer height the cross-section becomes higher and less wide, while the shape of the individual layers get more round. A reduced cross-sectional height leads to a reduced hydrostatic pressure, which results in a smaller relative deformation when $P_H > \tau_{printed}$. At larger $\tau_{printed}$ values, reducing the layer height also leads to lower D_B (cf. Fig. 10), which is due to the complex interplay between extrusion pressure, shape of the strands, and the contact area between layers. This complex interplay can be interpreted in the following way: when layers have a smaller height the extrusion pressure increases (see Fig. 10-b), which promotes larger deformation. However, the applied load of the deposited material is not exerted on the same contact area between layers (see Fig. 9) and in the cases with higher h, the contact area is smaller, which as an isolated effect leads to larger deformation. The latter effect is dominating in this case as seen in Fig. 10-a.

In Fig. 10-a, it is also seen that the relative deformation is less than 1 % for $h = 12.5$ mm and 18.75 mm when $\tau_{printed} = 500$ Pa and 700 Pa, respectively. This illustrates that by decreasing the layer height, the requirements for the yield stress buildup when aiming for geometrically stable layers is not as high, which as previously mentioned can be beneficial for some materials as a slower yield stress buildup can reduce crack formation in concrete as well as reduce the risk of residual stresses

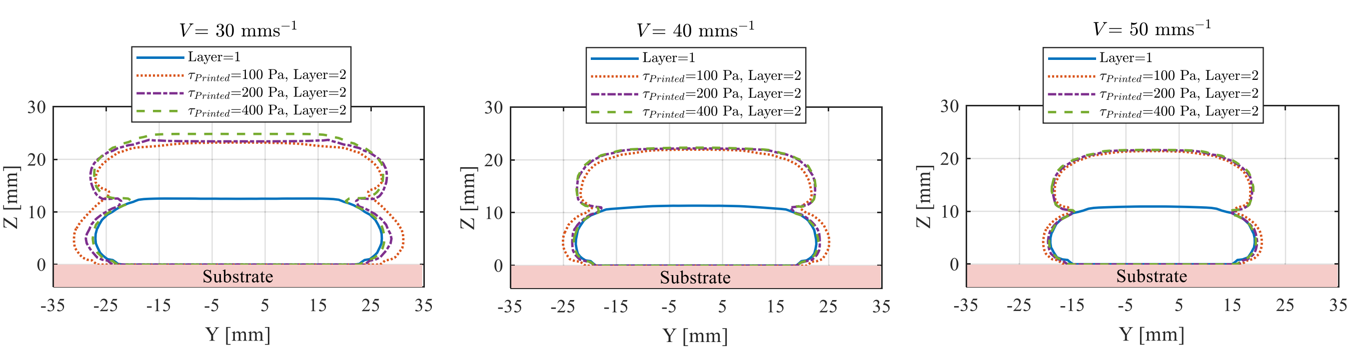

Fig. 5. Cross-sections of deposited layers for different printing speed V and printed yield stress $\tau_{printed}$.

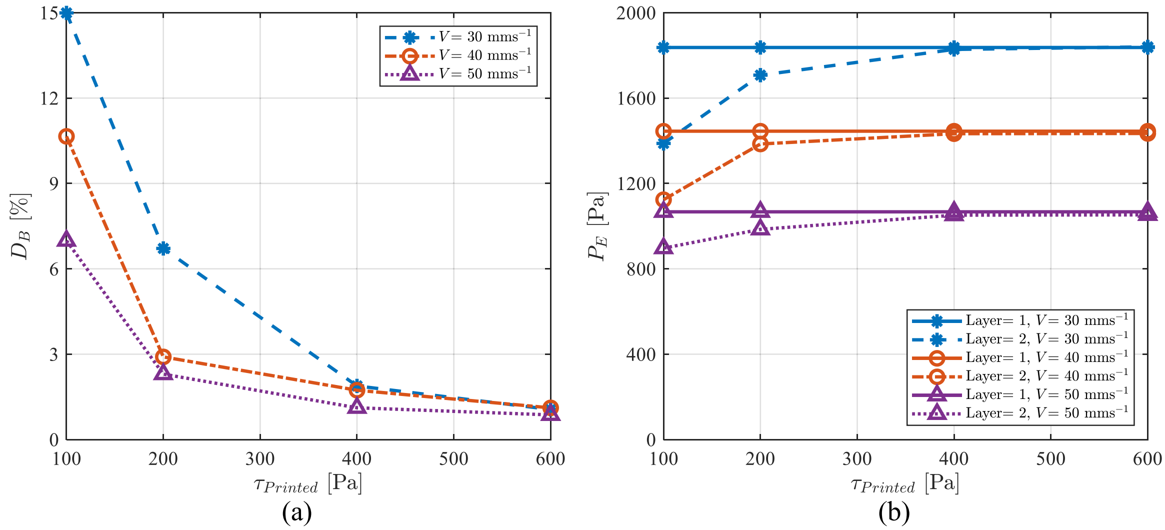

Fig. 6. (a) Relative deformation of first layer D_B and (b) extrusion pressure P_E as a function of printed yield stress $\tau_{printed}$, for different printing speed V.

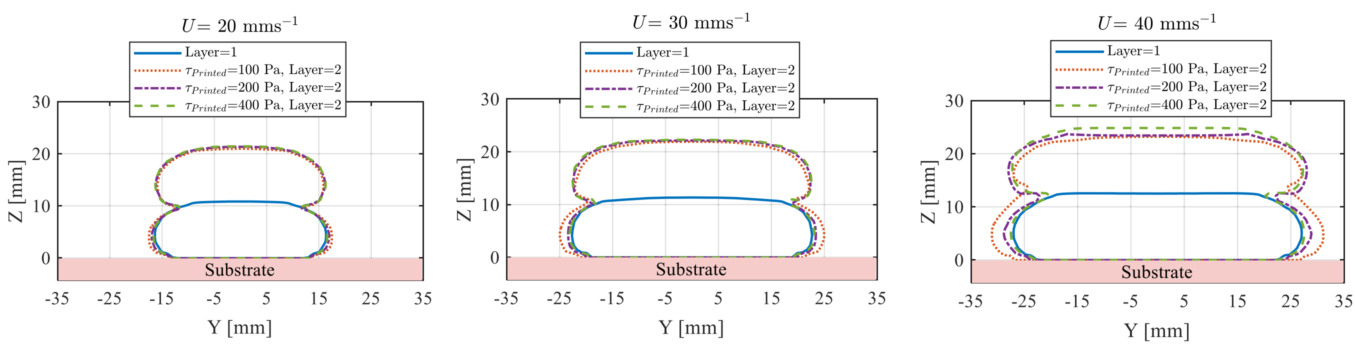

Fig. 7. Cross-sections of deposited layers for different extrusion speed U and printed yield stress $\tau_{printed}$.

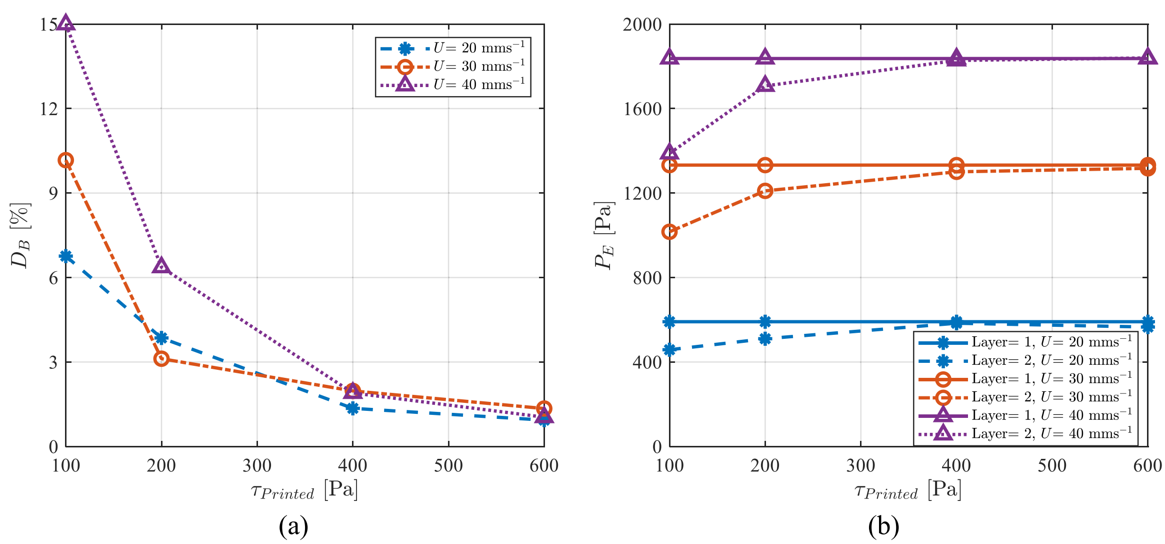

Fig. 8. (a) Relative deformation of first layer D_B and (b) extrusion pressure P_E as a function of printed yield stress $\tau_{printed}$, for different extrusion speed U.

and thermal warpage in thermosets. The rounder shape in the case of $h = 25$ mm, creates a situation where the relative deformation is ~1.8 % at $\tau_{printed} = 700$ Pa, which can be unsatisfactory for many application that require geometrical precision. As conclusion, rounder shapes with smaller interlayer contact area require a higher yield stress buildup to control the deformation.

3.2.4. Nozzle diameter

Different nozzle diameters are varied to investigate the geometrical stability in different printing scales. Larger nozzle diameters, $D = 25$ mm (Case ID 1), and $D = 50$ mm (Case ID 10) are used in big area additive manufacturing (typically 3DCP), whereas smaller nozzle diameters, $D = 0.4$ mm (Case ID 8), $D = 1.5$ mm (Case ID 9) are used in

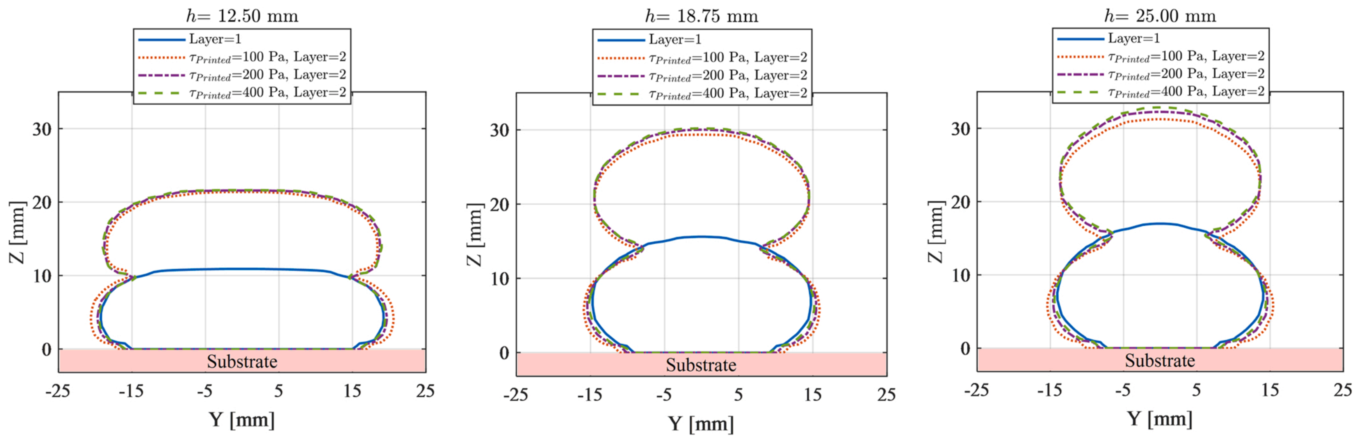

Fig. 9. Cross-sections of deposited layers for different layer height h and printed yield stress $\tau_{printed}$.

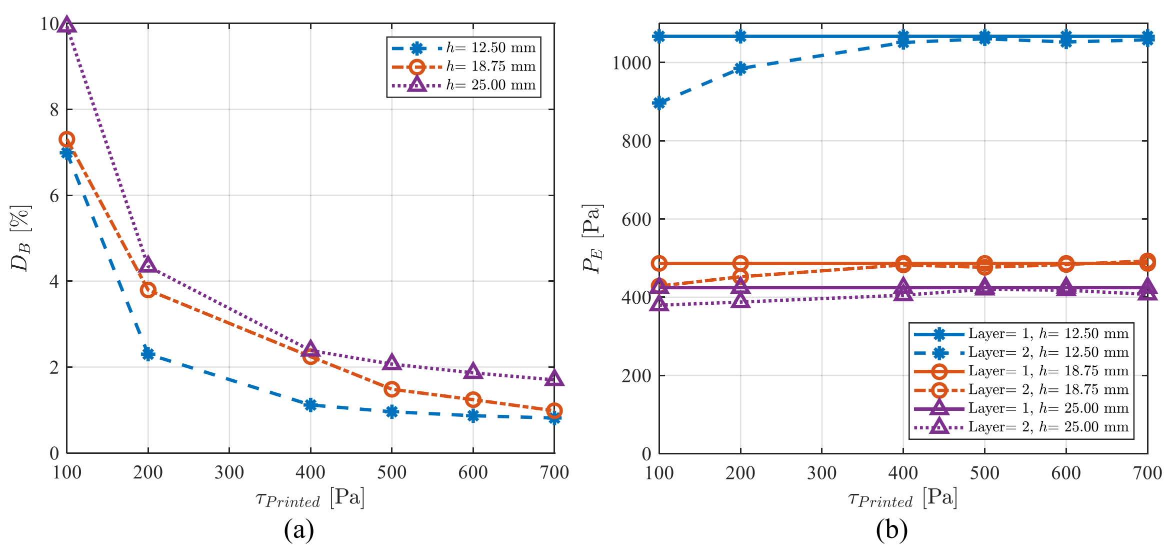

Fig. 10. (a) Relative deformation of first layer D_B and (b) extrusion pressure P_E as a function of printed yield stress $\tau_{printed}$, for different layer height h.

RC. In these simulations $U = 40$ mm/s, $V = 50$ mm/s, and $\eta_P = 33$ Pa.s. Fig. 11 compares the cross-sections of deposited layers for different printed yield stresses. Note that the axes are non-dimensionalized by the nozzle diameter. The largest relative deformation is seen for the smallest nozzle diameter, while the smallest relative deformation is observed for $D = 25$ mm. This non-monotonic behavior between relative deformation and nozzle diameter was also observed and discussed in [33], although in that study the minimum was found at $D = 1.5$ mm, which is attributed to the difference in printing yield stress applied in the two studies. Later in this section, more information about the effect of materials' rheology on geometrical stability will be addressed for different D. In Fig. 12-a, it is seen that for the two smallest nozzle diameters only limited improvement in D_B is obtained by increasing $\tau_{printed}$, which is due to the very high extrusion pressure (Fig. 12-b). In fact, it seems that the extrusion pressure increases by one order of magnitude when the nozzle diameter is reduced by one order of magnitude for these rheological parameters ($\tau_{printing} = 100$ Pa and $\eta_P = 33$ Pa.s). The high extrusion pressure for $D = 0.4$ mm forces the second layer to penetrate much more into the first layer, as compared to $D = 25$ mm where the second layer flows more horizontal (see Fig. 11).

The extrusion pressure is highly dependent on the rheological behavior of the material during printing as seen in Fig. 13. For large nozzle diameters, the extrusion pressure can be reduced by decreasing $\tau_{printing}$, but of course it cannot be lower than the hydrostatic pressure, as this would result in unwanted deformations. The plastic viscosity dominates the extrusion pressure for small nozzle diameters. This is due to the fact that most of the material inside the nozzle is yielded when the diameter is so small, and hence the value of the yield stress does not play a significant role. At a plastic viscosity of 0.05 Pa.s, the extrusion pressure is lowest but different in magnitudes for the two smallest nozzle diameters. In addition, the hydrostatic pressure is substantially lower for the small nozzle diameters, which reduces the yield stress buildup requirement. Consequently, geometrically stable prints can be obtained with the two smallest nozzle diameters with a limited yield stress buildup as long as the plastic viscosity is low enough, as seen in Fig. 14, which depict examples of stable printing scenarios (<1 % deformation) for all the investigated nozzle diameters. This illustrates that the rheological behavior of the deposited material requires special attention when aiming for stable printing.

4. Conclusions

A computational fluid dynamics model was used to predict the stability of deformable layers during MEX-AM of viscoplastic materials. The model applied a scalar approach to alter the viscous behavior of the fluid within the layers. The fluid was modelled using two Bingham constitutive models with different yield stress to represent the printing of wet material onto a semisolid printed layer. The cross-sections of deposited layers, relative first layer deformation, as well as extrusion

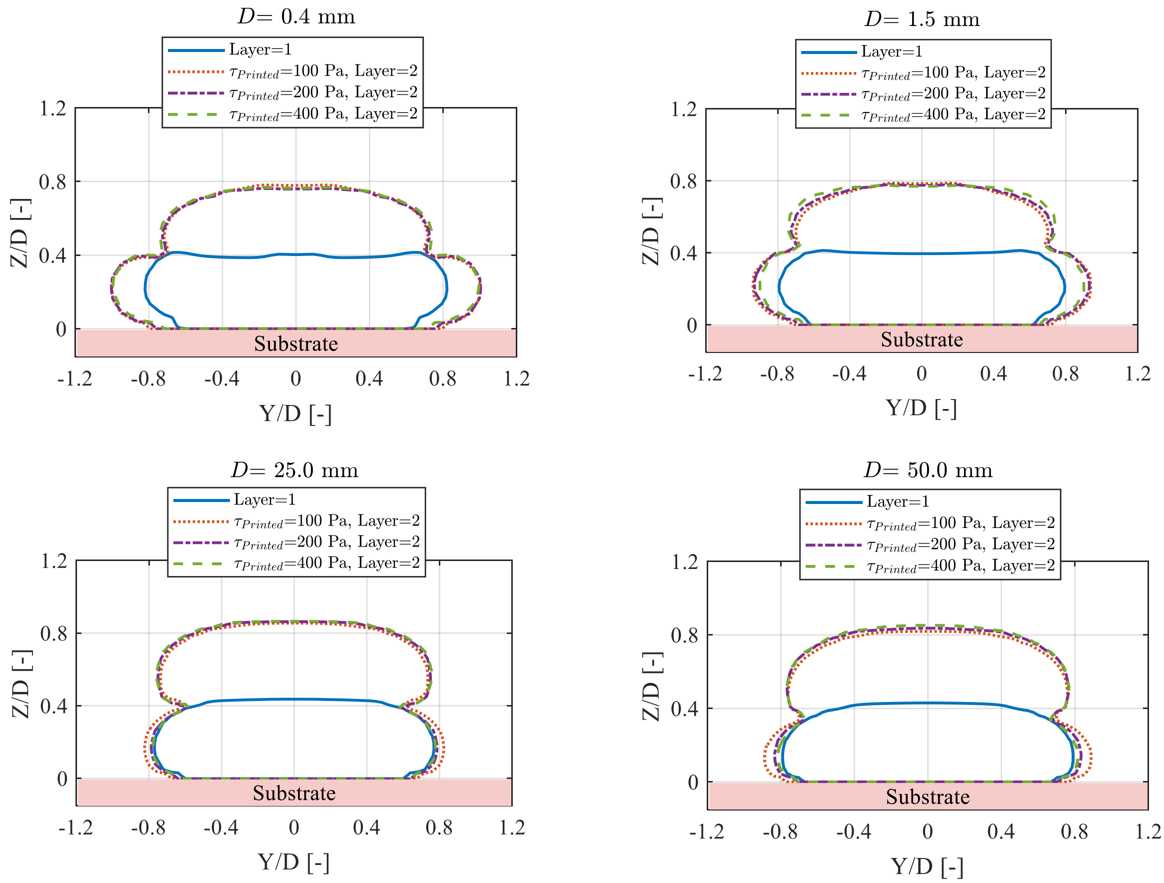

Fig. 11. Cross-sections of deposited layers for different nozzle diameter D and printed yield stress $\tau_{printed}$.

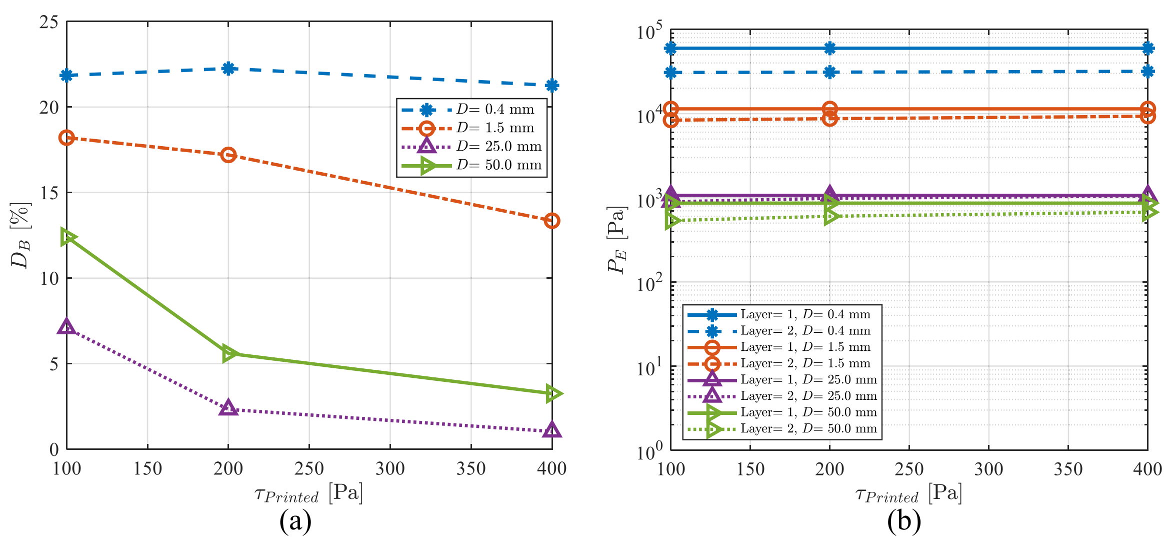

Fig. 12. (a) Relative deformation of first layer D_B and (b) extrusion pressure P_E as a function of printed yield stress $\tau_{printed}$, for different nozzle diameter D.

pressure were investigated for different yield stress buildup of the already printed material and for different processing conditions. It was found that when the yield stress of the printed layer is higher, the printing of a new layer produces less deformation than what can be seen for no yield stress buildup (i.e., wet-on-wet print). This is of course attributed to the higher effective viscosity of the first layer when printing wet-on-semisolid. It was also found that the increase in yield stress buildup reduced the deformation in a non-linear manner and that the largest improvement was obtained from the initial increase in yield stress.

When increasing the printing speed or reducing the extrusion speed, the study showed that less material is deposited, which leads to less extrusion- and hydrostatic-pressure and thereby less deformation. The combined effect of printing/extrusion speed and yield stress buildup has the largest influence on the speeds that leads to more material deposition, which is ascribed to the larger deformation observed at low yield stresses where the hydrostatic pressure exceeds it. Nevertheless, the yield stress at which geometrically stable prints were observed (i.e., a

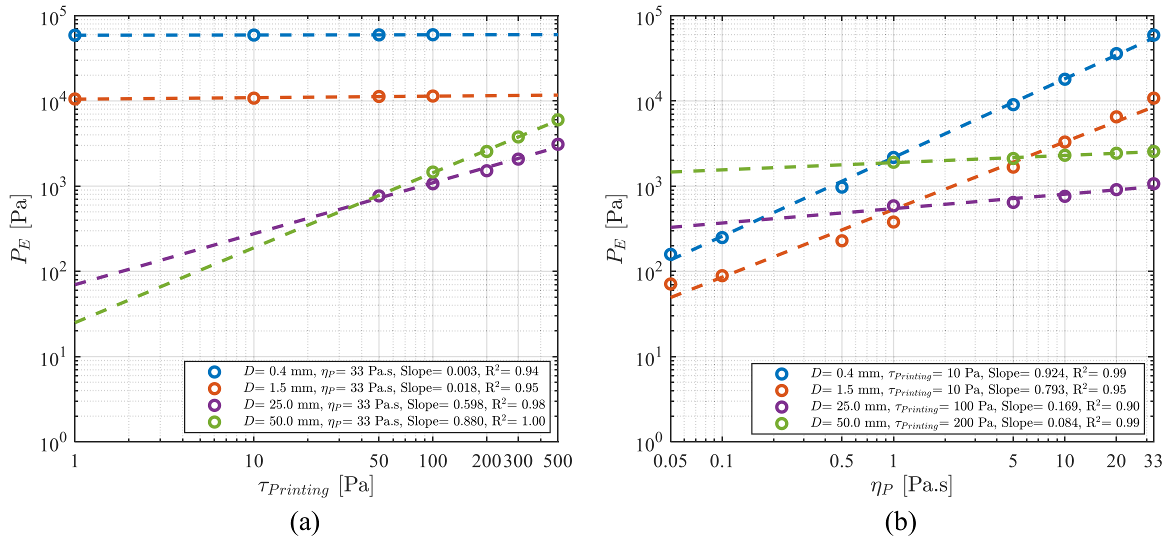

Fig. 13. Extrusion pressure P_E for different nozzle diameters D, (a) as a function of the printing yield stress $\tau_{printing}$ and (b) as a function of the plastic viscosity η_P. Note that $U = 40$ mm/s and $V = 50$ mm/s.

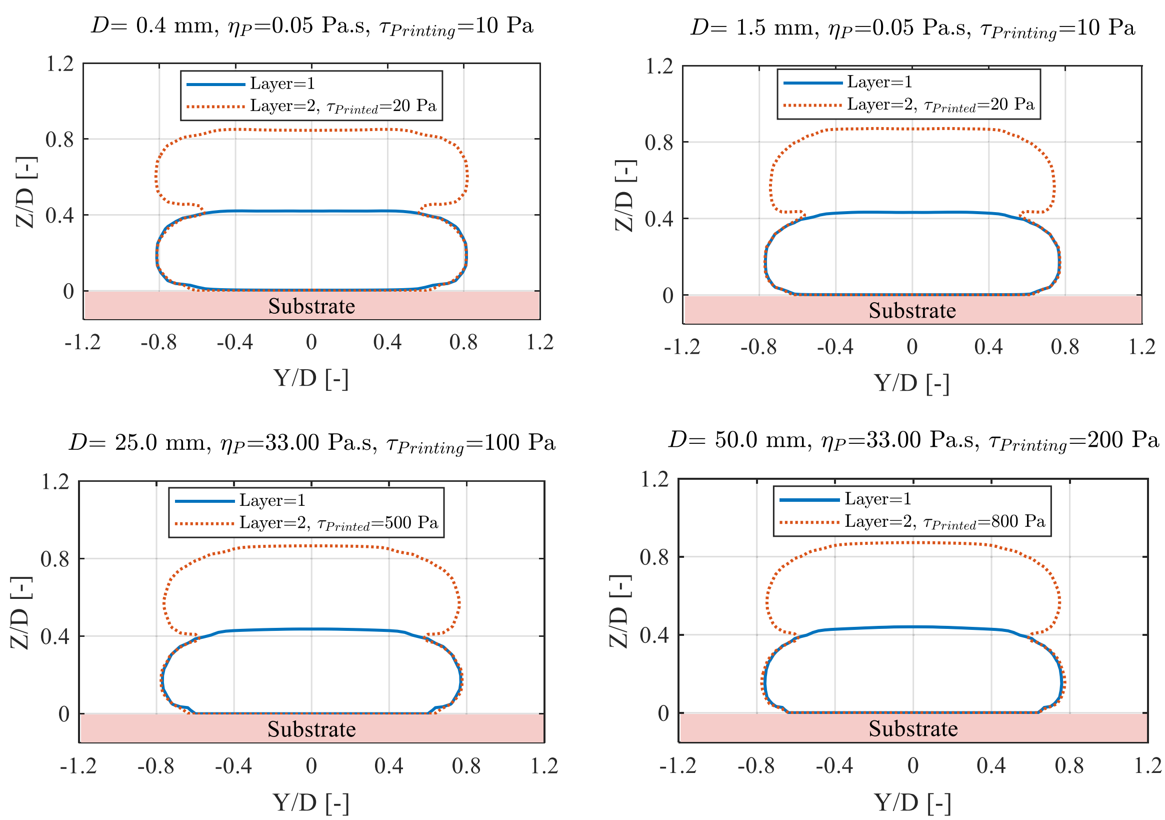

Fig. 14. Stable printing scenario for different nozzle diameters D. Note that $U = 40$ mm/s, and $V = 50$ mm/s.

relative deformation of ~1 %) was the same for all investigated printing- and extrusion-speeds indicating the limited gain by varying these two parameters at least when combined with the process parameters applied in this study. An interesting finding was though that the speed ratio (i.e., the printing speed divided by the extrusion speed) and the yield stress buildup did not show a monotonic behavior, which indicated that each speed had an independent effect on the deformation. This was attributed to the complex interplay between extrusion pressure, shape of the strands, and the contact area between layers, which especially was pronounced when varying the layer height. By increasing the layer height, a more round-shaped cross-section was formed, which led to a smaller contact area between the layers where the load from the deposited layer was distributed on. Consequently, a larger relative deformation was observed when increasing the layer height even though such increase reduced the extrusion pressure. In regards to time dependent material properties, it meant that stable prints could be obtained at a lower yield stress buildup when printing with a reduced layer height, which can be beneficial for some materials as a too quick material development can result in crack formation and warpage.

For all the investigated nozzle diameters, it was possible to obtain geometrically stable prints when taking into account a time-dependent yield stress. For the small nozzles, the extrusion pressure was substantially reduced by decreasing the plastic viscosity of the material and thus stable prints could be obtained by a limited yield stress buildup. For the larger nozzles, conversely, the extrusion pressure is dominated by the yield stress of the printing material, but this property could not be reduced too much as it needed to be higher than the hydrostatic pressure in order not to result in unwanted deformations. Therefore, larger nozzles required a larger yield stress buildup (both in absolute and relative terms) to produce stable prints.

In future works, the developed CFD model could be extended to include non-isothermal printing as some materials generate heat when solidifying/curing, which can affect the rheological behavior of the material and thereby the stability of the prints. The model could also be used to investigate the effect of a time-dependent plastic viscosity, which seems to be important for especially small nozzle diameters.

CRediT authorship contribution statement

Spangenberg Jon: Writing – review & editing, Supervision, Resources, Project administration, Investigation, Formal analysis, Conceptualization. **Šeta Berin:** Writing – review & editing, Investigation, Formal analysis. **Serdeczny Marcin P.:** Writing – review & editing, Investigation, Formal analysis. **Comminal Raphaël:** Writing – review & editing, Supervision, Formal analysis, Conceptualization. **Mollah Md Tusher:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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